

A Multi-Architecture Deep Learning Approach to Gold Price Forecasting Using Technical and Macroeconomic Indicators

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Abstract

This paper presents a high-capacity deep learning framework for daily gold log-return prediction. Instead of treating gold as a purely price-based time series, the final pipeline combines price-derived indicators, wavelet-denoised components, cross-asset returns, macro-financial variables, HMM regime indicators, and a sentiment/risk proxy. Eight architectures are implemented and compared: TFT-Lite, Residual-LSTM, CNN-GRU, CNN-BiLSTM, Temporal Convolutional Transformer (TCT), Multi-Scale Fusion (MSF), Informer, and Autoformer. Models are trained using an expanding-window walk-forward validation on daily data from January 2007 to May 2026. Feature selection uses a correlation, variance inflation factor, and MI+RF pipeline, reducing 176 cleaned candidate inputs to 30 selected features. The framework also includes split conformal prediction for uncertainty calibration and an Adaptive Regime-Conditioned Ensemble (ARCE) that conditions model weights on HMM-detected regimes. MSF achieves the strongest empirical test-set performance, while ARCE provides competitive accuracy with lower drawdown. The conformal intervals achieve approximately 90.2% empirical coverage against a 90% target. The study implements a reproducible forecasting pipeline that combines state-of-the-art architectures with interpretable feature selection, regime detection, uncertainty quantification, and trading-oriented validation.

Keywords: Gold price forecasting, Deep Learning, LSTM, CNN-GRU, CNN-BiLSTM, Temporal Fusion Transformer, Informer, Autoformer, Multi-Scale Fusion, Hidden Markov Model, Feature selection, Macroeconomic indicators, Sentiment proxy.

1. Introduction

1.1 Background

Gold plays an important role in global finance. However, the prediction of its price is still relatively difficult. The price change does not follow the same rules as stocks or bonds, and its value can fluctuate for a lot of different reasons. Its price rises when people are anxious, such as during war or economic trouble periods. Other times it goes up when inflation is high, or when the dollar is weak. And then there are times when it increases because of problems between countries. All these factors can be happening at the same time, and they can pull gold's price in different directions. This makes it really hard to predict what will happen to gold's price next, because it doesn't just depend on one thing. It's like trying to solve a puzzle with many moving pieces.

Gold is also regarded as a safe haven asset and an important reserve asset of global financial markets, especially during the economic uncertainty periods. Its price movements are influenced by a joint effect of various factors like macroeconomic conditions, central bank movements, financial market variables, and investor sentiment [14][18]. Other factors such as interest rates, inflation, central bank policies, currency movements, market volatility, and behavioral responses to global events all contribute to changes in gold prices. As a result, forecasting gold prices in an accurate way remains challenging. It is valuable to improve the predication accuracy for researchers, investors, portfolio managers, and policymakers.

Traditional financial time-series forecasting methods like ARIMA and GARCH are widely used to model price movements and volatility [16]. Although they provide useful benchmarks, their ability to capture non-linear relationships, multi-scale dependencies, and structural changes in financial data is rather limited [3]. Machine learning methods such as Support Vector Machines (SVM) and Artificial Neural Networks (ANN) improve non-linear modeling, but many still lack temporal depth or depend heavily on manually selected input sets [3].

Recent advances in deep learning have introduced architectures such as LSTM, CNN-LSTM hybrids, and Transformer-based models. Those architectures have shown strong potential in financial forecasting tasks [4][16][19]. LSTM networks capture temporal dependencies, CNN-LSTM models combine local pattern extraction with sequence learning, and Transformer architectures capture global dependencies through attention mechanisms [16][17][19][20]. However, much of the existing literature still relies mostly on price-volume-based inputs and its variants technical indicators and treats external variables such as sentiment and macroeconomic factors only as secondary features [18].

Most existing forecasting models treat gold price prediction as a purely price-volume-based problem. They put historical gold prices into statistical models or machine-learning algorithms, sometimes adding a few technical indicators such as RSI or moving averages. While these approaches have had some success, they fail to consider other important factors that also affect gold prices: variables such as news sentiment, geopolitical risk, the U.S. Dollar Index, real interest rates, and market fear gauges such as the VIX are not just correlated with gold; they often provide additional information beyond the price-volume data [1][14][18]. Some of them provide leading information, while others have long-term influence on gold price movements.

1.2 Research Objectives and Hypotheses

This paper proposes a framework that addresses this gap. We build a forecasting pipeline that integrates a broad candidate universe including price and volume derived indicators, macroeconomic variables, and sentiment/risk proxies and then applies a systematic data-driven selection process to identify the inputs that really related to

movements. Rather than relying on a single model, we train eight deep learning architectures in parallel, trying to capture a different aspect of gold prices evolution and combine their predictions through an ensemble that adapts its model weights to the current market regime.

The study does not assume that gold prices can be predicted perfectly. Instead, it tries to improve the out-of-sample forecasting relative to simple benchmarks using a combination of feature engineering, feature selection, walk-forward validation, regime detection, and uncertainty calibration. The final input uses daily data from January 2007 to May 2026 and implements 8 architectures: TFT-Lite, Residual-LSTM, CNN-GRU, CNN-BiLSTM, TCT, MSF, Informer, and Autoformer.

The study is also fueled by the thought that not all relevant information can be digested and reflected instantaneously. Central-bank purchases are disclosed with a delay, geopolitical risk may be priced with a lag, and structural demand from non-speculative buyers can cause behavioral biases in commodity markets. These frictions do not imply that the market can be beaten across all horizons, but they provide a theoretical basis for testing whether macroeconomic-related and investment sentiment-related indicators contain additional information not yet fully reflected in prices.

The study raises four empirical hypotheses. H1: The proposed deep learning models achieve statistically meaningful directional predictability relative to simple baselines. H2: adding macro and sentiment variables improves performance relative to price-volume-only features. H3: explicit indicators generated by HMM regime detection add marginal forecasting value beyond raw input features. H4: Split conformal prediction intervals can achieve approximately 90% empirical coverage on the held-out test set. These hypotheses are examined through model-level performance metrics, feature-group validation, HMM ablation, and conformal coverage diagnostics.

2. Theoretical Framework

2.1 Foundational Models and Financial Time-Series Prediction

Financial time-series prediction is not easy because asset returns are often noisy, nonlinear, heteroscedastic, and unstable across regimes. Classical models such as ARCH and GARCH are useful for volatility modeling, while ARIMA provides a transparent linear benchmark. However, these traditions often fail to deliver good results when the forecasting task is nonlinear and requires long temporal memory [3][16]. Deep learning models provide a better solution for those problems by learning nonlinear representations directly from sequential data [4][6].

Early work focused on LSTM networks, which are effective in capturing temporal dependencies in financial time series and have been widely applied to asset price prediction [4][6]. New LSTM variants have been developed to improve gate structures, enhance learning efficiency, and reduce redundancy, as well as to deal with limitations such as overfitting and computational complexity [5].

2.2 LSTM and Recurrent Architectures

LSTM was initially introduced by Hochreiter and Schmidhuber and then became a standard base model for sequence learning tasks [23]. Long Short-Term Memory networks are specialized recurrent neural networks. They are designed to capture long-term dependencies and mitigate the vanishing-gradient problem [4].

The standard function of LSTM cell can be described as follows:

$$\begin{aligned}
f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f), \\
i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i), \\
\tilde{C}_t &= \tanh(W_C[h_{t-1}, x_t] + b_C), \\
C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \\
o_t &= \sigma(W_o[h_{t-1}, x_t] + b_o), \\
h_t &= o_t \odot \tanh(C_t).
\end{aligned} \tag{1}$$

- f_t , i_t , and o_t are the forget, input, and output gates;
- C_t and $\tilde{C}_t$ are the cell and candidate cell states;
- h_t is the hidden state; x_t is the current input vector;
- $\sigma(\cdot)$ is the logistic activation function;
- $\odot$ denotes element-wise multiplication.

These gating equations regulate information flow and help LSTMs mitigate the vanishing-gradient problem [5].

In this project, recurrent modeling appears in several architectures: Residual-LSTM, CNN-GRU, CNN-BiLSTM, and the recurrent components of TFT-Lite and MSF. The Residual-LSTM architecture learns from ResNet-style design and pairs causal convolutions with skip connections across stacked LSTM layers. CNN-GRU and CNN-BiLSTM combine causal convolutional layers with recurrent blocks, so local price and volatility patterns are extracted before temporal dependencies are modeled [10][16].

2.3 Architectural Evolution of CNN-LSTM Hybrid Models

The CNN-LSTM hybrid framework has pointed an important direction in gold price prediction because it combines local feature extraction with sequential learning [16][17]. The studies of Livieris et al. (2020) and Santika et al. (2021) have laid the foundation for combining CNN. They used a feature extractor with LSTM as a sequence learner. These hybrid frameworks can capture both short-term and long-term dependencies, and they outperform isolated CNN or LSTM baselines [16][17].

The introduction of signal decomposition methods to extract the signal from the noise is a further innovation, which results in higher accuracy and robustness [15]. Liang et al. (2022) used the ICEEMDAN module to decompose the original signal into intrinsic mode functions, then applied LSTM, CNN, and a convolutional block attention module to jointly forecast all sublayers [15]. However, the computational latency of running independent joint forecasting networks for each decomposed layer can make such systems difficult to implement in agile live-trading environments.

Bidirectional temporal processing is another research direction. Amini and Kalantari (2024) implemented a CNN-BiLSTM hybrid framework to process sequential data in both forward and backward chronological directions [10]. Other recent studies succeeded in adding attention-augmented layers beyond the basic CNN-LSTM framework. He et al. (2019) integrated attention into an LSTM-CNN model, while Mohamed et al. (2025) used CNN-LSTM-AM with BiLSTM for short-term gold forecasting [12][13]. These studies prove the importance of the use of hybrid architectures but often retain limited inputs, single-split validation, or weaker statistical testing.

2.4 Transformer Architectures and Attention Mechanisms

Transformer-based models have become trending approaches because the attention mechanisms they use can model global relationships across multiple input features. The Temporal Fusion Transformer combines recurrent layers for local processing with attention mechanisms for global dependencies and interpretable variable weighting [19]. Informer improves efficiency using ProbSparse attention, enabling scalable modeling of

long time series [20]. Autoformer introduces time-series decomposition and replaces standard attention with auto-correlation to improve long-term forecasting stability [21].

Transformer-based NLP models such as FinBERT have also been combined with financial forecasting frameworks to incorporate news sentiment into prediction tasks [22]. In this study, Informer and Autoformer are included not as the only focus but as modern state-of-the-art benchmarks against recurrent, convolutional-recurrent, and multi-scale architectures.

2.5 Multivariate Feature Fusion and Behavioral Indicators

Input data selection and feature engineering play an important role in gold price prediction. Early studies often only used historical closing prices and volume as the input or added price-based technical indicators such as moving averages, RSI, and MACD [10][16][17]. Recent research extended the range by incorporating financial market and macroeconomic variables. Taneva-Angelova et al. (2025) proposed a multi-indicator framework containing financial market indicators, macroeconomic variables, and sentiment indicators [18].

Behavioral factors such as news and sentiment have been also taken into consideration. Ji and Wen (2026) integrated media sentiment analysis through natural language processing and proposed a CNN-QRLSTM model to analyse panic-driven volatility and improve directional accuracy [14]. These innovations show a shift of research direction toward multimodal deep learning. The deep learning models integrate heterogeneous data such as historical prices, macroeconomic indicators, and sentiment signals, making it better deal with volatility and complex interdependencies [1][14].

Table 1. Literature Comparison Table

Study	Architecture	Dataset	Key Inputs	Validation	Stat. Tests
Livieris et al. (2020)	CNN-LSTM	10 yr	Gold price only	Single split	-
Amini & Kalantari (2024)	CNN-BiLSTM + grid search	44 yr	Gold price only	Single split	-
Taneva-Angelova et al. (2025)	10 architectures	2014-2024	Price, macro, sentiment	Single 80/20	-
Ji & Wen (2026)	CNN-QRLSTM + NLP	n/a	Price + NLP sentiment	Single split	-
He et al. (2019)	LSTM-CNN + attention	Short sample	Price-based TA	Single split	-
Mohamed et al. (2025)	CNN-LSTM-AM + BiLSTM	n/a	Price-based TA	Single split	-
This study	8 DL models + ARCE	2007-2026	Price, macro, cross-asset, regime, sentiment/risk proxy	Walk-forward + holdout	Binomial, DM, PT, HAC Sharpe

2.6 Research Gap and Proposed Intervention

The research gap is not simply the lack of another deep model. The gap is the absence of a compact and reproducible framework that jointly handles feature overload, model diversity, regime change, uncertainty intervals, and economically meaningful evaluation. Current quantitative implementations for gold price prediction continue to face the same set of problems: too many correlated features, point forecasts without uncertainty, static ensembles, reliance on price alone, non-stationarity assumption and regime change, reproducibility risk, and high training cost.

The first problem is feature overload. Although modern data infrastructure makes it easy to compute hundreds of technical indicators and macroeconomic series, adding more features does not always help. Many indicators are mathematically correlated, and multicollinearity increases variance without adding new information. Manual pre-selection introduces bias, while indiscriminate inclusion leads to overfitting and long training times.

The second problem is the previous results give point forecasts without uncertainty. A trader needs to know not only what the model prediction is but also how wide its plausible

range is. The third problem is the usage of static ensembles: the relative skill of an attention-based model, a recurrent model, or a convolutional model is not constant across market conditions. The fourth problem is reliance on price-volume data alone, which emphasizes what the market has already done over the leading signals that give the hints of what it will do next.

This study contributes to the literature in four ways. First, it moves beyond a price-only model by adding macro-financial and behavioral variables as economically relevant indicators rather than merely auxiliary inputs. Second, it uses a controlled 3-step feature-selection funnel to reduce redundancy while preserving interpretability. Third, it compares eight architectures and introduces regime-detection ensembling through ARCE. Fourth, it adds uncertainty calibration through split conformal prediction and evaluates the system with walk-forward validation, statistical tests, and trading-oriented metrics.

3. Methodology

3.1 Data Collection and Multi-Source Framework

The project is organized as a full architecture that takes raw market data on one end and produces calibrated gold-return forecasts, prediction intervals, and a diagnostic dashboard on the other. The pipeline is broken into a numbered notebook with 19 sections, each of which can be read on its own but works with the others.

The final pipeline constructs a combined daily dataset from January 2007 to May 2026. The raw market data contains 4,840 daily rows and 17 base series. A helper function downloads daily series for gold, silver, copper, crude oil, the GDX miners ETF, the S&P 500, the U.S. Dollar Index, EUR/USD, USD/JPY, the VIX, Treasury yields, and the Federal Funds Rate via API. Sources include yfinance and FRED, and all series are forward-aligned on a common business-day calendar.

The data layer also combines macroeconomic and behavioral inputs, including TIPS 10-year real yield, breakeven inflation, CPI, M2, high-yield credit spreads, financial stress, the World Uncertainty Index, the Geopolitical Risk Index, COT gold positioning, and World Gold Council central-bank reserve data. These variables are included because gold is affected by currency movements, real-rate expectations, commodity demand, risk appetite, and volatility conditions [18]. All non-daily time-series are adjusted also using the forward-alignment method.

Table 2. Data Source Categories

Data block	Examples in the final pipeline	Purpose
Market and cross-asset prices	Gold futures, silver, copper, crude oil, GDX, S&P 500, U.S. Dollar Index, EUR/USD, USD/JPY, VIX, Treasury yields	Captures currency movements, commodity demand, risk appetite, volatility conditions, and cross-asset spillovers.
Financial and macro indicators	FRED series, Fed Funds Rate, TIPS 10-year real yield, breakeven inflation, CPI, M2, high-yield spreads, financial stress, WUI, and GPR	Encodes real-rate conditions, monetary policy, inflation pressure, credit stress, uncertainty, and geopolitical risk.
Market microstructure and positioning	Gold futures volume, roll-gap indicators, COT gold positioning, and World Gold Council central-bank reserve data	Adds information about trading activity, positioning, liquidity, and structural reserve demand where reliable data are available.
Behavioral and risk proxy layer	FinBERT/VADER design where headlines are available; otherwise VIX, RSI, S&P 500 movements, Fear-and-Greed style variables, and related moving averages	Approximates sentiment and market fear when a full historical news archive is unavailable.

3.2 Sentiment and Behavioral Proxy

An NLP sentiment layer is contained in our implementation. The intended primary source is FinBERT, a finance-tuned BERT model, applied to financial headlines. Headlines are converted into a sentiment score using positive-minus-negative probabilities, and daily scores are aggregated so that days with more news coverage carry more information [22].

The practical difficulty is that free RSS endpoints retain only a limited history of headlines. In the final run, RSS headline scraping did not provide a usable historical headline archive. Therefore, the sentiment block should be interpreted cautiously as a sentiment/risk proxy rather than a full historical FinBERT news archive. Where real news sentiment is unavailable, the pipeline falls back to market-risk information such as VIX behavior, RSI, and related moving averages. A production deployment could replace this layer with Bloomberg, Refinitiv, FactSet, RavenPack, or another historical news database without changing the rest of the architecture.

3.3 Wavelet Denoising and Target Construction

The target variable for all models is the daily log return of gold, defined as the natural logarithm of the ratio of consecutive closing prices. Log returns are used for three reasons. First, raw gold prices are normally non-stationary, while log returns provide a more stable approximation for neural-network training. Second, log returns are usually more symmetric than raw percentage returns, which helps stabilize gradient computation during training. Third, they are additive across time, making a 60-day lookback window interpretable as accumulated market information.

The pipeline applies a Daubechies-4 wavelet decomposition to the gold close series and retains trend and detail components as additional features. The denoised log-return series is also retained as an alternative input signal. This allows the model to separate high-frequency market noise from broader directional information while still preserving the original return target.

3.4 Exhaustive Feature Engineering and Selection

The feature-engineering module is designed to be exhaustive intentionally. It allows any indicator that seems plausible to be considered and then uses data to make the selection rather than relying on the analysts intuition instead. The module creates 181 total columns before cleaning; a pool of 176 features is left after removing sparse and all-NaN columns.

The features are divided into several categories. Features derived from price data include momentum indicators like RSI, MACD, Stochastic Oscillator, Williams %R, CCI, MFI, and short-window momentum; trend indicators such as EMA/SMA horizons, Bollinger Bands, Donchian Channels, ADX, and price-position measures; and volatility features such as rolling standard deviations, realized variance, ATR, volatility ratios, and volatility-regime indicators. Other transformations like lagged returns, persistence measures, wavelet-based signals, denoised-return features, calendar effects, and futures roll-related indicators are also included.

Beyond price-related variables, the feature-engineering module also transforms cross-asset, macro-financial, geopolitical, policy, and behavioral inputs into signals that could better fit the model. Returns, changes, momentum terms, z-scores, spreads, growth rates, and regime indicators derived from the U.S. Dollar Index, VIX, S&P 500, oil, copper, silver, GDX, DBC, TIP, Treasury yields, TIPS real yields, breakeven inflation, Fed Funds Rate, high-yield credit spreads, financial stress, M2, CPI, WUI, GPR, COT positioning, central bank reserve and purchase data, and the sentiment/risk proxy are added. This design incorporates not only price-volume behavioral, but also real-rate expectations, inflations, currency movements, commodity demand, risk appetite,

geopolitical uncertainty, policy demand, and behavioral stress before the downstream selection pipeline determines which signals are informative.

What sets this approach apart is not the list of features per se, but the disciplined three-stage funnel that follows. In Step A, the pairwise absolute correlation matrix is computed, and one feature from each pair with an absolute correlation higher than 0.75 is eliminated, keeping the feature with higher mutual information with the target. Step B uses the Variance Inflation Factor test and iteratively eliminates variables until all survivors have $VIF < 5.0$. Step C ranks the rest of the features with a combined mutual-information and random-forest importance score. The final input width is not ex ante fixed. Instead, K is adaptively chosen from $\{15, 20, 25, 30\}$ by validation MAE.

Table 3. Realized Feature-Selection Output in the Final Notebook Run

Stage	Rule	Remaining Features	Key Output
Raw usable pool	After sparse-column cleaning	176	Full engineered candidate set
Step A	Correlation filter $ \rho > 0.75$	99	Higher-MI feature retained within each pair
Step B	VIF threshold < 5.0	87	Multicollinearity reduced
Step C	MI + Random Forest rank, adaptive K	30	Selected input width in this run

The adaptive funnel chose 30 inputs for the K model in the last notebook run. This number should be interpreted as the realized output of the final run, not a fixed rule: if the candidate pool, validation period, random-forest ranking, or candidate K values change, the selected width and individual selected variables may also change. At the category level, the chosen set is still economically interpretable, with wavelet components, short-window volatilities and momentum indicators, currency variables, commodity-basket returns and real-rate or credit-spread proxies featuring prominently.

3.5 Feature-Group Experimental Design

We also builds three lightweight experimental feature groups to test whether additional information sources improve forecasting performance, in addition to the main selected-input pipeline. Features are classified by source data, not by subjective economic interpretation. Derived features inherit the group of their source. For example, a rolling z-score of GPR is considered macro since GPR is a macro/geopolitical input, while gold volatility is price-based since it is calculated using only gold price data.

Group 1 includes technical information based on prices and volumes, such as lagged returns, volatility measures, momentum variables, wavelet components and calendar controls. In Group 2, we add macro-financial and cross-asset variables, such as USD returns, Treasury and TIPS variables, high-yield spreads, broad commodity returns, and equity market signals. Group 3 adds available sentiment and behavioral proxy variables that survive cleaning and group-specific selection, including the NLP/risk proxy layer and usable positioning, fear/risk, or reserve-related variables where available.

The comparison from G1 to G2 to G3 separates two empirical questions: whether macro and cross-asset information improves on price-only forecasting, and whether the additional sentiment/behavioral proxy layer adds incremental value beyond the macro feature set. In the final version of implementation, this grouping is used as a lightweight validation block rather than as the main eight-model experiment; therefore, its feature counts should not be confused with the adaptively selected input width used in the primary model comparison.

3.6 Market Regime Detection

To account for structural nonstationarity, the framework fits a three-state Gaussian Hidden Markov Model on the training partition only. The final framework uses three regimes, then fits the HMM under multiple covariance structures (diagonal, tied, and spherical) and five random starting points. The specification with the highest training log-likelihood is retained; if no HMM converges successfully, the system falls back to K-means clustering. The observation vector includes gold log returns, 21-day realized volatility, RSI-14, VIX, USD returns, and a GPR proxy. Fitting the regime model only on the training partition prevents look-ahead bias.

The fitted regimes are added directly to the feature matrix and also serve a second purpose: they are the key that the regime-conditioned ensemble uses to choose model weights. In other words, the HMM is not only a descriptive state classifier; it also supplies the conditioning variable for ARCE.



Figure 1. Gold Price, HMM Market Regimes, and Daily Log Returns Across the 2007-2026 Sample.

3.7 Custom Loss Functions and Training Controls

A model trained only to minimize MSE may not get the direction right. We defines a custom loss functions, which includes a Huber loss that penalizes wrong directional predictions, a sign-agreement loss that mixes MSE with binary cross-entropy on the sign, and an adaptive asymmetric loss for long- or short-biased strategies. These losses are used to shape models toward directional skill while remaining robust to outliers in financial returns.

Shared training utilities include Adam with gradient clipping, early stopping with best-weight restoration, and a ReduceLROnPlateau callback. Dropout and validation monitoring are used to reduce overfitting. All scalers, feature-selection steps, and regime labels are fitted on training data only.

3.8 Deep Learning Architectures

The final experiment compares eight deep learning architectures. Each architecture reflects a different inductive bias: attention over sequence history, residual recurrence, convolutional-recurrent hybridization, explicit multi-scale modeling, or efficient long-sequence attention. A comparison is designed to test whether model diversity adds robustness across regimes, since no single architecture can capture everything.

Table 4. Model Architecture Summary

Model	Core Structure	Main Inductive Bias
TFT-Lite	Projection + multi-head attention + LSTM	Attention-weighted temporal learning
Residual-LSTM	Causal CNN stem + stacked residual LSTM	Long memory with gradient highways
CNN-GRU	Causal CNN + bidirectional GRU	Local extraction + gated recurrence
CNN-BiLSTM	Causal CNN + bidirectional LSTM	Symmetric temporal context
TCT	Dilated TCN + self-attention + dual head	Parallel receptive field + direction-aware output
MSF	Parallel LSTM branches at 20/60/120 days	Explicit multi-horizon fusion
Informer	ProbSparse attention	Efficient long-sequence Transformer
Autoformer	Decomposition + auto-correlation	Trend/seasonality decomposition

The Temporal Convolutional Transformer combines dilated temporal convolutional blocks with self-attention and a dual regression/direction output head. This gives the model a multi-scale receptive field and allows the directional head to regularize the regression output. Multi-Scale Fusion takes a different approach: it runs parallel LSTM branches at 20-, 60-, and 120-day horizons and combines them through a learned softmax gate. The network can therefore up-weight short horizons during choppy markets and longer horizons during trending periods without manual switching logic.

3.9 Walk-Forward Training and Disk Caching

The final selected feature matrix is converted into 60-day sliding-window sequences. The chronological split of 75% / 15% / 15% is used, which produces 3,345 training sequences, 716 validation sequences, and 718 test sequences. The MSF model uses additional parallel 20-, 60-, and 120-day lookback branches internally. This design is used to prevent new information from leaking backward into the training process.

One limitation of a single chronological train/validation/test split lies in that although it gives one estimate of out-of-sample performance, it fails to show enough about how performance varies across time. Our study implements expanding-window walk-forward validation with three folds. Each fold trains on all data up to the current point and validates on the next period, mimicking the production setting where a model is periodically retrained on accumulated history.

To reduce training cost, a disk-cache layer is added. After each model finishes walk-forward training, its weights, walk-forward metrics, and training history are persisted to disk. On later runs, models with intact cache files can be loaded instead of retrained. This makes iterative methodology development practical when adding or changing a model architecture.

3.10 Adaptive Regime-Conditioned Ensemble and Conformal Prediction

The proposed ARCE learns one ensemble weight vector per HMM regime using validation-set predictions rather than averaging the predictions of the base models with fixed weights. For each regime, the validation actual returns are regressed on the matrix of model predictions, and the resulting weights are normalized. At inference time, the HMM emits a regime label for each test day, and the corresponding weight vector is applied.

Table 5. ARCE Learned Weights by Regime

Regime	TFT	Res-LSTM	CNN-GRU	CNN-BiLSTM	TCT	MSF	Informer	Autoformer
0	0.000	0.078	0.000	0.341	0.102	0.479	0.000	0.000
1	0.000	0.047	0.003	0.073	0.174	0.475	0.227	0.000
2	0.125	0.125	0.125	0.125	0.125	0.125	0.125	0.125

The uncertainty layer uses split conformal prediction. On the validation set, the system computes the absolute residual for every observation; the 90% residual quantile becomes the conformal half-width. At inference, the prediction interval is formed by adding and subtracting this calibrated half-width from each forecast. This is attractive because it is

distribution-free, does not assume normally distributed residuals, and requires no extra model training.

3.11 Evaluation Metrics and Statistical Tests

A common weakness in gold-forecasting literature is that papers report only RMSE and directional accuracy and ignore economically meaningful measures of skill. The notebook implements a broad evaluation suite covering regression accuracy, directional accuracy, ranking quality, and risk-adjusted strategy returns. It also runs statistical tests to distinguish genuine forecast skill from noise.

For economic interpretation, the predicted log-return sequence is also transformed back into an implied gold-price path by cumulative exponentiation from the last observed price. This reconstruction is used for visual diagnostics and price-path interpretation, while the primary statistical evaluation remains at the daily log-return level.

Table 6. Evaluation Protocol

Metric/Test	What It Measures	Why It Matters
RMSE / MAE	Magnitude accuracy	Absolute forecast error in log-return units
R2	Variance explained	Calibration of prediction scale
DirAcc	Sign agreement	Core trading utility
Large-move DirAcc	Sign on large moves	Tail-risk and large-signal usefulness
IC Pearson / Spearman	Rank correlation	Factor-research quality
Sharpe / MaxDD / Calmar	Risk-adjusted strategy behavior	Portfolio relevance
Binomial / PT tests	Directional skill vs chance	Statistical validation of sign prediction
Diebold-Mariano	Forecast comparison	Tests improvement over benchmark forecasts

4. Results

4.1 Walk-Forward Cross-Validation

The walk-forward validation confirms that the strongest architectures are the ones that combine either multi-scale sequence processing or attention-based learning. MSF has the lowest mean walk-forward RMSE and the highest mean directional accuracy. Informer and TFT-Lite also perform strongly, while Autoformer is weaker in this daily return setting.

Table 7. Walk-Forward Summary Across Three Folds

Model	Mean RMSE	Std RMSE	Mean DirAcc	Std DirAcc
MSF	0.4813	0.0574	82.95%	1.72%
Informer	0.5595	0.0583	81.91%	0.47%
TFT-Lite	0.6264	0.0397	80.74%	0.60%
TCT	0.6521	0.1227	77.83%	2.39%
CNN-BiLSTM	0.7575	0.0979	74.18%	5.42%
Residual-LSTM	0.7131	0.1423	71.47%	2.76%
CNN-GRU	0.8142	0.0939	69.68%	1.35%
Autoformer	0.9422	0.1033	62.37%	1.62%

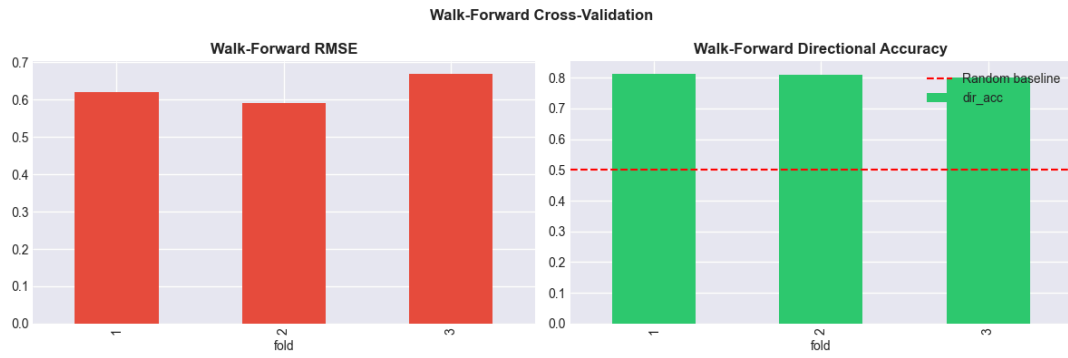


Figure 2. Walk-Forward Cross-Validation Summary for RMSE and Directional Accuracy.

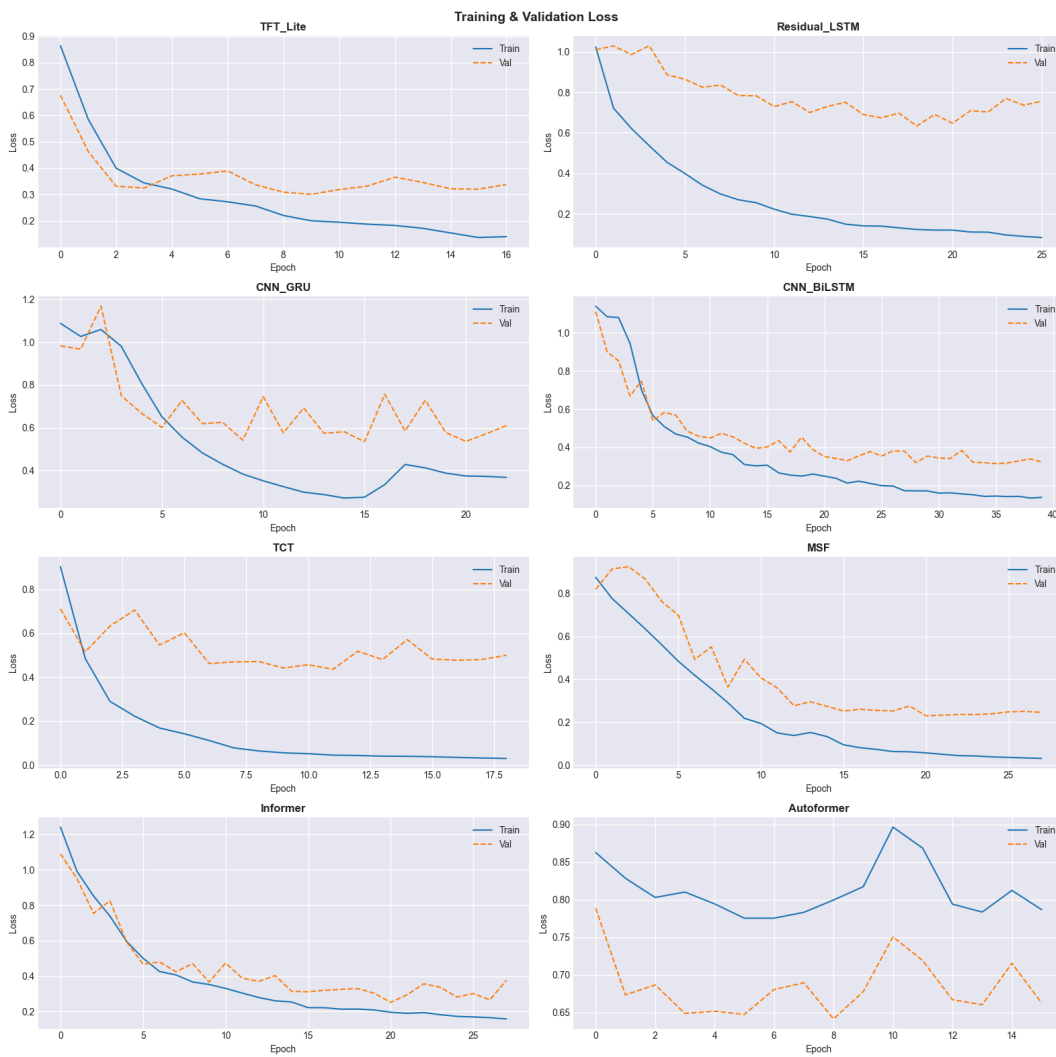


Figure 3. Training and Validation Loss Curves for the Eight Evaluated Architectures.

4.2 Final Test-Set Model Performance

On the held-out test set, MSF is the best model by RMSE, directional accuracy, and Sharpe ratio. It achieves an RMSE of 0.006460 and a directional accuracy of 84.82%. ARCE ranks second in directional accuracy at 83.57% and has the lowest maximum drawdown among the main strategies, indicating that regime-conditioned averaging improves risk control even when the best single architecture remains MSF.

Table 8. Final Test-Set Performance Summary

Model	RMSE	MAE	R2	DirAcc	Large DirAcc	Sharpe	MaxDD
MSF	0.006460	0.003969	0.7389	84.82%	98.89%	12.211	-2.93%
ARCE	0.008029	0.004491	0.5967	83.57%	98.89%	12.057	-2.26%
Informer	0.008083	0.004723	0.5913	81.34%	96.67%	10.853	-3.91%
TFT-Lite	0.008548	0.005207	0.5430	80.08%	95.00%	10.732	-3.91%
CNN-BiLSTM	0.008724	0.005414	0.5239	78.83%	95.00%	10.459	-2.93%
TCT	0.009177	0.005785	0.4731	78.55%	96.67%	10.544	-3.91%
Res-LSTM	0.010171	0.006771	0.3529	71.45%	86.67%	7.578	-9.91%
CNN-GRU	0.010525	0.007047	0.3071	67.69%	85.00%	6.998	-5.84%
Autoformer	0.012646	0.008580	-0.0003	56.41%	58.89%	1.447	-19.52%
Persistence	0.018346	0.012824	-1.1055	48.33%	46.11%	-0.769	-57.78%

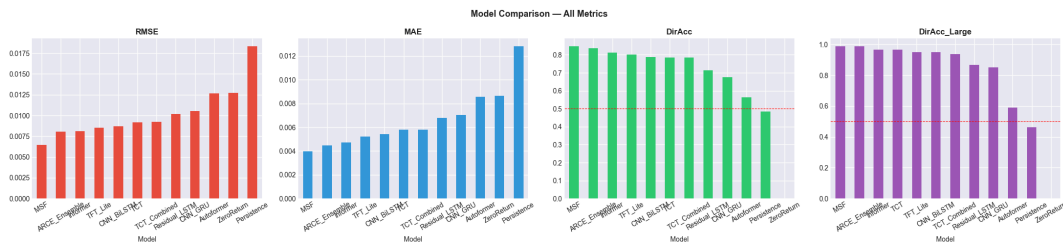


Figure 4. Final Model Comparison Across RMSE, MAE, Directional Accuracy, and Sharpe Ratio.

The gap between MSF and the recurrent-only models suggests that gold return prediction benefits from explicit multi-scale learning. The 20/60/120-day structure allows the model to combine short-term momentum, medium-term trend behavior, and longer quarterly market structure. This is consistent with the economic nature of gold, which reacts both to immediate risk shocks and slower macro-financial repricing.

4.3 Conformal Prediction Intervals

The conformal module reaches 90.2% empirical coverage against a 90% target. Validation coverage is 90.36%, and the average interval width in return space is 0.012689. Regime-conditional calibration produces slightly wider intervals in more uncertain states while preserving the target coverage. This result is important because the model output is not only a point forecast but also a calibrated uncertainty range.

Table 9. Split Conformal Prediction Results

Metric	Value
Target coverage	90%
Validation coverage	90.36%
Test coverage	90.2%
Global $\hat{q}$ -hat (scaled residual space)	0.570956
Average interval width (return space)	0.012689

Note: $\hat{q}$ -hat is reported in the scaled residual space used by the conformal calibration, while interval width is shown after transformation back to return space.

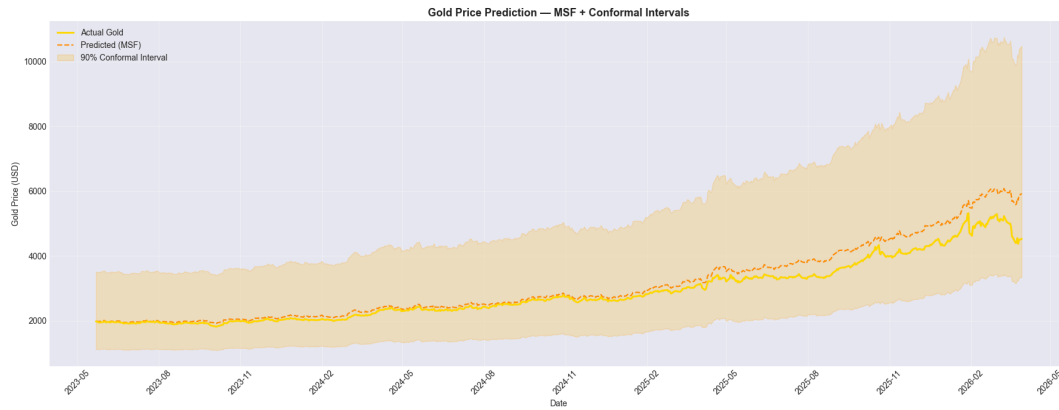


Figure 5. Conformal Prediction Intervals Around the Gold Price Forecast Path.

4.4 Feature-Group Validation

The lightweight feature-group validation uses the Residual-LSTM model to compare price/technical features, price plus macro/cross-asset features, and price plus macro plus sentiment/behavioral features. G2, the price plus macro/cross-asset set, performs best with RMSE of 0.010674 and directional accuracy of 77.30%. This supports H2 in the lightweight validation design: macro and cross-asset information improves forecasting relative to price-only features.

Table 10. Lightweight Feature-Group Validation

Feature Group	N	RMSE	MAE	R2	DirAcc	Large DirAcc	Sharpe
G1	40	0.011052	0.007268	0.2309	70.29%	85.16%	7.188
Price/Technical	40	0.010674	0.006985	0.2825	77.30%	91.76%	9.219
G2 Price + Macro	40	0.011656	0.007723	0.1445	64.92%	81.32%	5.348
G3 Price + Macro + Sentiment							

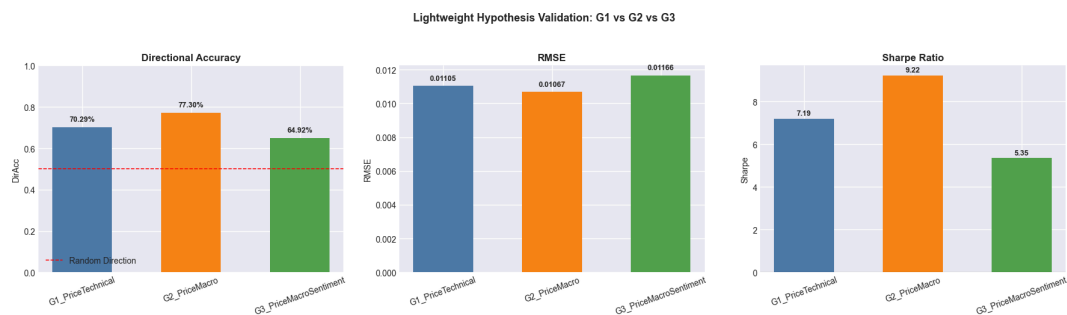


Figure 6. Lightweight Feature-Group Validation for G1, G2, and G3.

The weaker G3 result should not be interpreted as proof that sentiment is irrelevant to gold. The final run did not have a complete archived headline dataset, and the sentiment layer was therefore largely proxy-driven. The empirical result is more conservative: with the current proxy design, adding sentiment/behavioral variables does not improve daily return prediction beyond the macro/cross-asset feature set.

4.5 HMM Ablation

The HMM ablation compares the final selected feature set with and without explicit regime indicators. In the lightweight validation block, adding HMM features improves RMSE from 0.010971 to 0.010490 and directional accuracy from 65.20% to 71.66%. The improvement in directional accuracy is about 6.46 percentage points. This supports the idea that regime labels can add marginal information, although the result should be interpreted as validation evidence rather than a full proof of causal market-state structure.

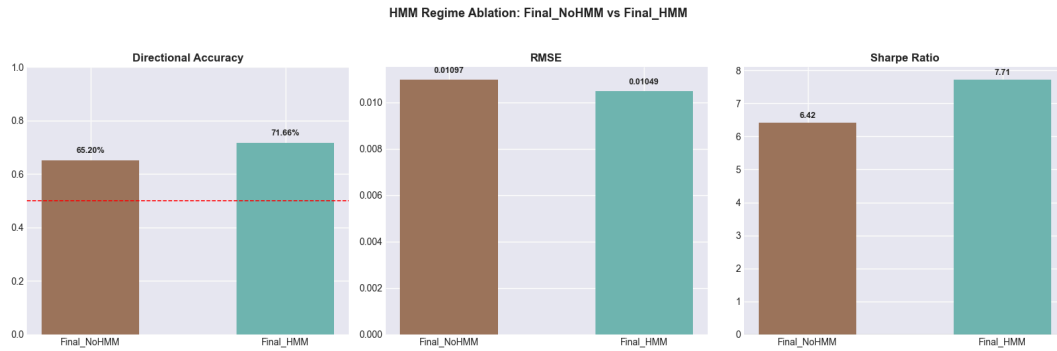


Figure 7. HMM Ablation Comparing Final_NoHMM and Final_HMM.

Table 11. Final NoHMM vs HMM Ablation

Model	N Features	RMSE	MAE	R2	DirAcc	Sharpe	Delta RMSE	Delta DirAcc
Final NoHMM	28	0.010971	0.007492	0.2420	65.20%	6.422	0.000000	0.00%
Final HMM	34	0.010490	0.006834	0.3070	71.66%	7.708	-0.000481	+6.46%

4.6 Statistical Tests and Backtest Interpretation

The statistical tests show that the main models have highly significant directional accuracy relative to a 50% random benchmark. MSF has a binomial p-value of $2.23\text{e-}85$ and a Pesaran-Timmermann statistic of 17.76. ARCE also shows strong directional skill, with a binomial p-value of $7.64\text{e-}79$ and a PT statistic of 16.77. Models such as Persistence fail to pass these directional tests.

Table 12. Selected Statistical Test Results

Model	N	DirAcc	Binom p	PT	PT p	Sharpe p
MSF	718	84.82%	$2.23\text{e-}85$	17.760	0.000	$3.59\text{e-}45$
ARCE Ensemble	718	83.57%	$7.64\text{e-}79$	16.773	0.000	$2.73\text{e-}48$
Informer	718	81.34%	$4.32\text{e-}68$	15.613	0.000	$1.15\text{e-}51$
TFT-Lite	718	80.08%	$1.70\text{e-}62$	14.693	0.000	$3.24\text{e-}46$
Persistence	718	48.33%	0.825	-1.306	0.192	0.896

The backtesting can be viewed as an indicative long/short signal test for study. It is not a deployable trading strategy since it excludes real trading-related cost considerations: transaction costs, bid-ask spreads, slippage, turnover controls, position-size constraints, and execution latency. Under this cost-free design, MSF and ARCE produce the strongest strategy results. MSF achieves the highest cumulative and annualized return, while ARCE achieves the best Calmar ratio because its maximum drawdown is the lowest among the main deep-learning strategies.

Table 13. Selected Backtest Results

Strategy	Cum Return	Ann Return	Ann Vol	Sharpe	MaxDD	Calmar
Buy & Hold	128.74%	33.70%	20.09%	1.678	-17.73%	1.90
MSF	25828.64%	603.36%	15.99%	37.743	-2.89%	208.92
ARCE Ensemble	24702.49%	592.48%	16.06%	36.889	-2.23%	265.64

Informer	17055.21%	508.44%	16.65%	30.538	-3.83%	132.72
TFT-Lite	16390.04%	500.05%	16.71%	29.928	-3.83%	130.53
Persistence	-35.65%	-14.33%	20.14%	-0.712	-43.89%	-0.33

Note: These figures are diagnostic signal-test results under a cost-free long/short assumption. They are not deployable trading results unless transaction costs, slippage, turnover constraints, volatility targeting, and position limits are added.

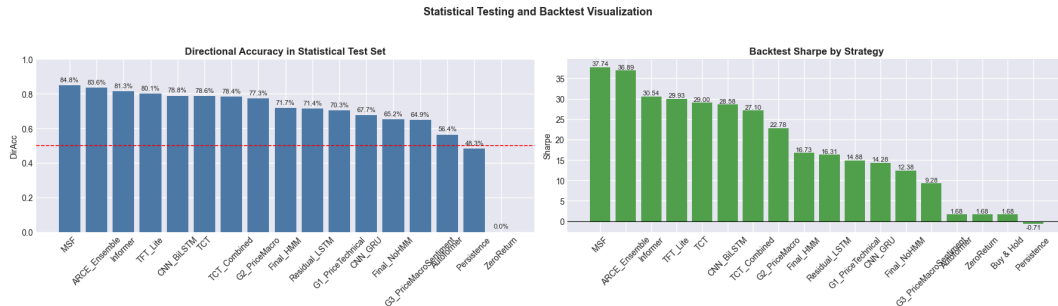


Figure 8. Statistical-Testing and Backtest Visualization From the Final Notebook.

5. Discussion

5.1 Interpretation of Empirical Findings

The results show that model architecture matters as much as feature breadth. MSF performs best because gold price movements contain information at several horizons: short-run momentum and volatility, medium-run trend behavior, and slower macro-financial repricing. ARCE is slightly weaker than MSF in raw forecast accuracy, but it is valuable from a risk-management perspective because it reduces maximum drawdown through regime-conditioned weighting.

The feature-group validation gives a cautious answer to the macro and sentiment question. The G2 feature set, which adds macro and cross-asset variables to price/technical signals, outperforms the price-only version. This supports the view that gold is not only a technical time series but also a macro-sensitive asset. However, the G3 sentiment layer performs worse in the current notebook run. This does not prove that sentiment is unimportant; it shows that a proxy-based sentiment layer is not strong enough to improve next-day return forecasting reliably.

The result also illustrates why daily return forecasting should be evaluated with more than one metric. R-squared values in financial return prediction can be modest even when directional skill is economically useful. In the final test set, the strongest models achieve high directional accuracy and favorable risk metrics, while the conformal layer gives an explicit uncertainty budget. This combination is more useful for portfolio management than a point forecast alone.

5.2 Project Contributions Relative to the Pain Points

Each of the pain points identified in the theoretical framework is addressed by a specific component of the project. Feature overload is solved by the correlation/VIF/MI+RF funnel. Introducing macro, cross-asset, wavelet, and sentiment/risk proxy features reduces the reliance on only price-volume data. Non-stationarity is addressed through HMM regime labels and regime-conditioned ensemble weights. Static ensembles are replaced by ARCE. Point forecasts without uncertainty are addressed through split conformal prediction. Reproducibility and training cost are managed through deterministic settings and disk caching.

Table 14. Pain-Point to Solution Map

Pain Point	Project Component
Feature overload	Three-stage MI-guided feature selection funnel
Reliance on price alone	Macro, cross-asset, wavelet, and sentiment/risk features
Non-stationarity and regime change	HMM regime labels and regime-conditioned intervals
Static ensembles	ARCE per-regime ensemble weights
Point forecasts without uncertainty	Split conformal prediction intervals
Reproducibility	Seed control, deterministic settings, no-shuffle fitting

5.3 Obstacles and Limitations in Gold Forecasting

Despite the improvements in the deep learning architecture, several fundamental obstacles limit the accuracy of gold price forecasting. Daily gold returns contain substantial random noise that no model can systematically learn, and sudden geopolitical events, central-bank decisions, or financial crises can move gold sharply in ways that do not recur in historical patterns.

Gold returns also show different patterns as markets and macroeconomic conditions change, so models trained on one regime may gradually lose relevance. High-capacity deep learning models further risk memorizing the training sample rather than learning relationships that generalize. Finally, extensive feature engineering across a fixed historical sample can capture chance patterns that do not persist out of sample. These limitations are why the study emphasizes walk-forward validation, held-out testing, conformal intervals, and future transaction-cost analysis rather than treating the backtest as deployable trading performance.

5.4 Practical Applications and Future Work

The framework has practical value for portfolio managers and quantitative researchers because it produces directional forecasts, uncertainty intervals, and model diagnostics rather than only a single point prediction. In a portfolio context, the signal could be used as a tactical overlay for gold exposure, a hedge-timing indicator, or a risk-management input when gold is used as a safe-haven asset.

The most important future improvement is the sentiment layer. The current notebook contains the structure for FinBERT/VADER-style sentiment scoring, but the final run did not have access to a complete historical headline archive. Because the sentiment data is mostly available from 2018. We used a robust proxy for a reasonable fallback if data are missing. Replacing the RSS-based layer with a paid or institutional news archive would allow the sentiment hypothesis to be tested more fairly.

A second extension is a realistic trading simulation. Future work should include transaction costs, bid-ask spreads, slippage, turnover constraints, volatility targeting, and position sizing. Without these elements, the backtest remains a diagnostic signal test rather than a production trading result.

A third extension is live deployment and model monitoring. The pipeline already has a modular structure. It can be converted into a production workflow that updates data daily, recalculates features, generates forecasts and conformal intervals, and monitors drift in model performance, feature distributions, and regime probabilities. Finally, the same architecture can be generalized to other liquid assets such as silver, oil, equity indices, FX pairs, and bond ETFs, although each target would require its own economic interpretation, indicators, and validation design.

6. Conclusion

This study develops a multi-source and multi-model framework for daily gold log-return forecasting. The final implementation combines 176 cleaned candidate features, a robust 3-step feature-selection funnel, eight neural-network architectures,

walk-forward validation, HMM regime detection, adaptive regime-conditioned ensembling, and split conformal prediction intervals.

The study experiments with the concept to identify the driving forces behind gold's price using a feature-ranking system with a feature-selection filter. Let the funnel select from a large set the top-ranked features that most influence the gold price. Surprisingly, some indicators, which we think otherwise won't be much correlated with the gold price, rank among the top. The concept can be replicated to discover other financial asset prices as well.

The main empirical finding is that the Multi-Scale Fusion model is the strongest architecture in the final test set. It achieves the best RMSE, directional accuracy, and Sharpe ratio, suggesting that explicit multi-horizon sequence modeling is well suited to gold. ARCE does not outperform MSF on raw accuracy, but it improves drawdown control and achieves the best Calmar ratio among the main strategies. The conformal module also performs as intended, reaching approximately 90% empirical coverage.

The feature-group validation provides a more nuanced conclusion. Price and technical variables remain highly informative at the daily horizon, but adding behavioral, macro, and cross-asset signals improves performance in the lightweight validation experiment.

The novelty of our work lies in shifting the objective from "what will happen" to "why it happens." The study seeks to move from black-box forecasting to a system that aligns statistical precision with economic meaning. By embedding the signals from selected features into a multi-scale architecture—interpretable feature selection, regime awareness, uncertainty calibration, and statistical validation—we can produce more robust forecasts than naive baselines. The strongest contribution of the study is the architecture: it turns a collection of models and indicators into a reproducible research pipeline that can be extended, audited, and eventually adapted to real portfolio-management workflows.

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Appendix

How the Pipeline Works - A Simple Walkthrough

Step 1 - Gathering the data. The pipeline collects daily gold data and related market, macroeconomic, cross-asset, and behavioral proxy variables from January 2007 to May 2026.

Step 2 - Cleaning and transforming the signal. The pipeline converts gold prices into log returns, generates technical indicators, adds wavelet-based components, and aligns all variables on a common business-day calendar.

Step 3 - Selecting the inputs. Instead of manually choosing variables, the pipeline uses correlation filtering, VIF filtering, and MI+RF ranking to reduce the feature space from 176 usable features to 30 final inputs.

Step 4 - Learning patterns with multiple architectures. Eight deep learning models analyze the selected features, each using a different inductive bias: recurrence, attention, convolution, multi-scale learning, or Transformer-style decomposition.

Step 5 - Quantifying uncertainty and risk. The final system produces point forecasts, conformal prediction intervals, regime-conditioned ensemble forecasts, and trading-oriented diagnostics.

Supplementary Architecture Details

TFT-Lite starts with a dense projection layer with 128 units and ELU activation, followed by dropout and multi-head attention with eight heads and a key dimension of 16. The attention output is added through a residual path and normalized; a feed-forward block then passes into two stacked LSTM layers with 64 and 32 units, followed by a 64-unit dense layer and a one-unit output. The model is compiled with Adam at a learning rate of 0.001, gradient clipping at unit norm, and the directional Huber loss.

Residual-LSTM uses residual connections directly in a recurrent framework to mitigate vanishing gradients. The model begins with a causal one-dimensional convolutional layer with 64 filters and batch normalization. Two stacked 128-unit LSTM blocks use dropout and skip connections, followed by layer normalization, a 64-unit LSTM, a 32-unit dense layer with L2 regularization and a linear output. The optimizer is Adam with a learning rate of 0.0005 and gradient clipping.

CNN-GRU combines causal convolutional layers and gated recurrent units to capture both short-term price patterns and longer dependencies. Two causal Conv1D layers with 64 filters extract local structure, followed by batch normalization, max-pooling, dropout, a bidirectional GRU layer with 128 units, a second GRU layer with 64 units, a 32-unit dense layer, and a linear output. The model uses Adam with a learning rate of 0.0008 and directional Huber loss.

CNN-BiLSTM follows the same convolutional-recurrent logic but replaces GRU layers with bidirectional LSTM layers. A causal Conv1D layer extracts local patterns, then batch normalization and max-pooling stabilize the representation. Two stacked bidirectional LSTM layers with 128 and 64 units model temporal context in both directions, with dropout between layers to reduce overfitting.

The Temporal Convolutional Transformer (TCT) combines a projection layer, dilated causal TCN blocks with dilation rates 1, 2, 4, and 8, multi-head self-attention, gated pooling, and a dual output head. The regression head predicts scaled log returns, while the auxiliary direction head is trained with binary cross-entropy. The combined loss gives 70% weight to regression and 30% to direction classification, so directional information regularizes the return forecast.

The Multi-Scale Fusion (MSF) network uses separate LSTM branches at 20-, 60-, and 120-day lookback horizons. Each branch applies a causal convolution followed by LSTM layers to produce a horizon-specific embedding. A learned softmax gate assigns per-sample weights to the three horizons before the gated sum is projected to the final output, allowing the network to emphasize short horizons in choppy markets and longer horizons in trending regimes.

Informer and Autoformer are included as modern Transformer-style benchmarks. The notebook implements an Informer-inspired model with positional encoding, attention blocks, and temporal distillation through convolution and pooling. The Autoformer-inspired model adds series decomposition after attention and feed-forward blocks to separate seasonal and trend components before final pooling and prediction.

Supplementary Notebook Diagnostics

The following appendix items preserve additional notebook outputs that are useful for auditability but too detailed for the main results section. They include the full model-level evaluation tables, statistical-test diagnostics, feature-redundancy visualization, and sentiment/risk proxy dashboards.

Table A1. Full Test Metrics: Forecast Accuracy and Directional Skill

Model	RMSE	MAE	R2	DirAcc	Large DirAcc	Rolling DirAcc	IC Spearman
MSF	0.006460	0.003969	0.7389	84.82%	98.89%	94.81%	0.8753
ARCE	0.008029	0.004491	0.5967	83.57%	98.89%	94.45%	0.8723
Informer	0.008083	0.004723	0.5913	81.34%	96.67%	91.48%	0.8355
TFT-Lite	0.008548	0.005207	0.5430	80.08%	95.00%	91.16%	0.8206
CNN-BiLSTM	0.008724	0.005414	0.5239	78.83%	95.00%	90.43%	0.7734
TCT	0.009177	0.005785	0.4731	78.55%	96.67%	90.66%	0.7696
TCT_Combined	0.009205	0.005796	0.4700	78.41%	93.89%	89.59%	0.7594
Residual-LSTM	0.010171	0.006771	0.3529	71.45%	86.67%	81.66%	0.6311
CNN-GRU	0.010525	0.007047	0.3071	67.69%	85.00%	79.64%	0.6837
Autoformer	0.012646	0.008580	-0.0003	56.41%	58.89%	56.67%	0.4193
Persistence	0.018346	0.012824	-1.1055	48.33%	46.11%	46.45%	-0.0601
ZeroReturn	0.012696	0.008638	-0.0083	0.00%	0.00%	0.00%	n/a

Table A2. Full Test Metrics: Strategy and Ranking Diagnostics

Model	Sharpe	MaxDD	Calmar	IC Pearson	Q5-Q1 Spread
MSF	12.211	-2.93%	189.66	0.8621	0.02913
ARCE	12.057	-2.26%	244.43	0.8049	0.02780
Informer	10.853	-3.91%	131.71	0.7977	0.02752
TFT-Lite	10.732	-3.91%	130.70	0.7747	0.02676
CNN-BiLSTM	10.459	-2.93%	171.14	0.7444	0.02568
TCT	10.544	-3.91%	129.12	0.7407	0.02558
TCT_Combined	10.150	-2.98%	164.89	0.7366	0.02559
Residual-LSTM	7.578	-9.91%	39.62	0.6284	0.02090
CNN-GRU	6.998	-5.84%	62.95	0.7156	0.02357
Autoformer	1.447	-19.52%	4.24	0.4641	0.01475
Persistence	-0.769	-57.78%	-0.76	-0.0529	-0.00383
ZeroReturn	-1.447	-107.42%	-0.77	n/a	n/a

Table A3. Full Statistical-Test Diagnostics

Model	N	Binom p	DM p vs MSF	PT	PT p	Sharpe p
MSF	718	2.23e-85	1.000	17.760	0.000	3.59e-45
ARCE	718	7.64e-79	0.0125	16.773	0.000	2.73e-48
Informer	718	4.32e-68	0.0061	15.613	0.000	1.15e-51
TFT-Lite	718	1.70e-62	0.0012	14.693	0.000	3.24e-46
CNN-BiLSTM	718	3.31e-57	0.0008	13.655	0.000	3.18e-41
TCT	718	4.52e-56	1.13e-05	13.319	0.000	1.06e-39
TCT_Combined	718	1.65e-55	8.47e-06	14.706	0.000	7.96e-37
Residual-LSTM	718	1.25e-31	3.49e-07	11.612	0.000	2.82e-36
CNN-GRU	718	6.93e-22	9.03e-09	5.801	6.58e-09	5.98e-25
Autoformer	718	3.37e-04	1.34e-09	0.000	1.000	2.91e-03
Persistence	718	0.825	1.20e-08	-1.306	0.192	0.896
ZeroReturn	718	1.000	9.71e-10	-16.657	0.000	2.91e-03

Supplementary Figures

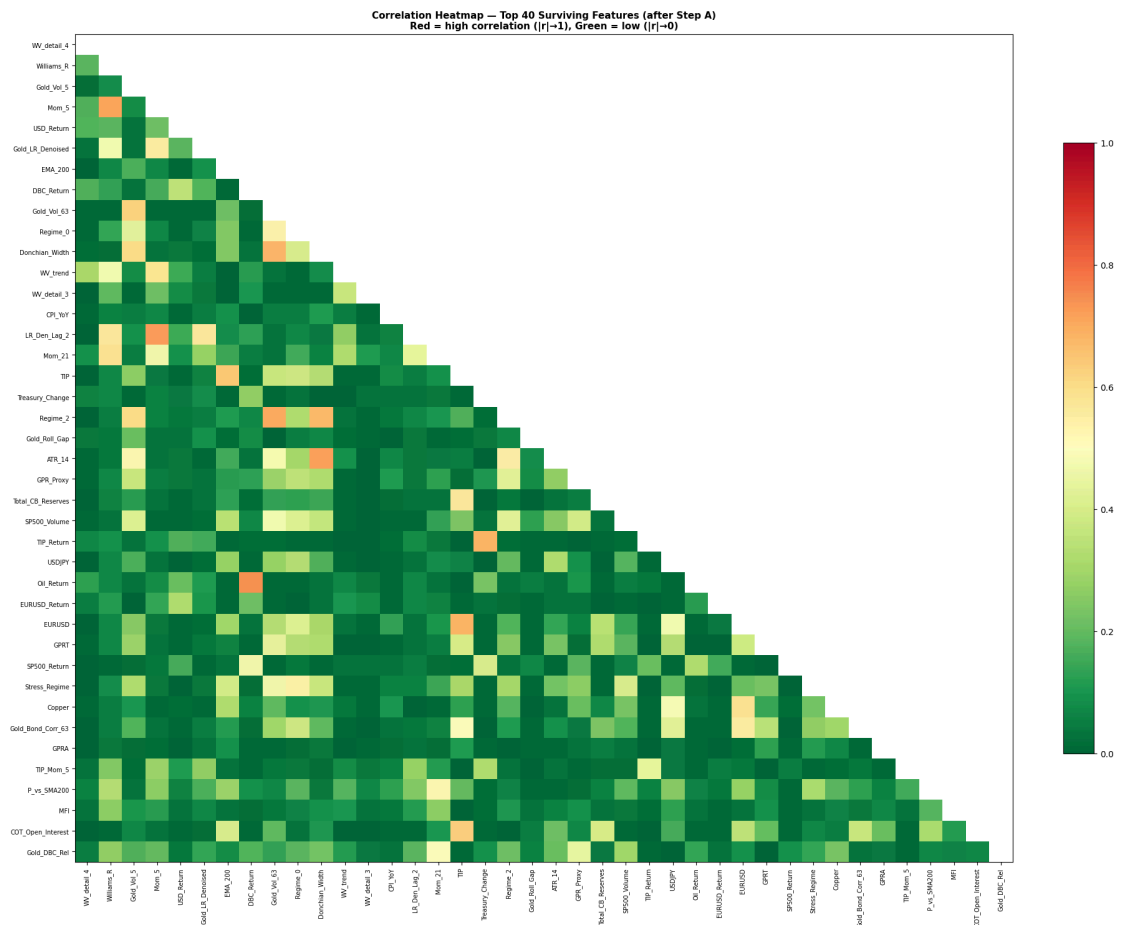


Figure 9. Feature Correlation Heatmap Used to Diagnose Redundancy Before Feature Selection.

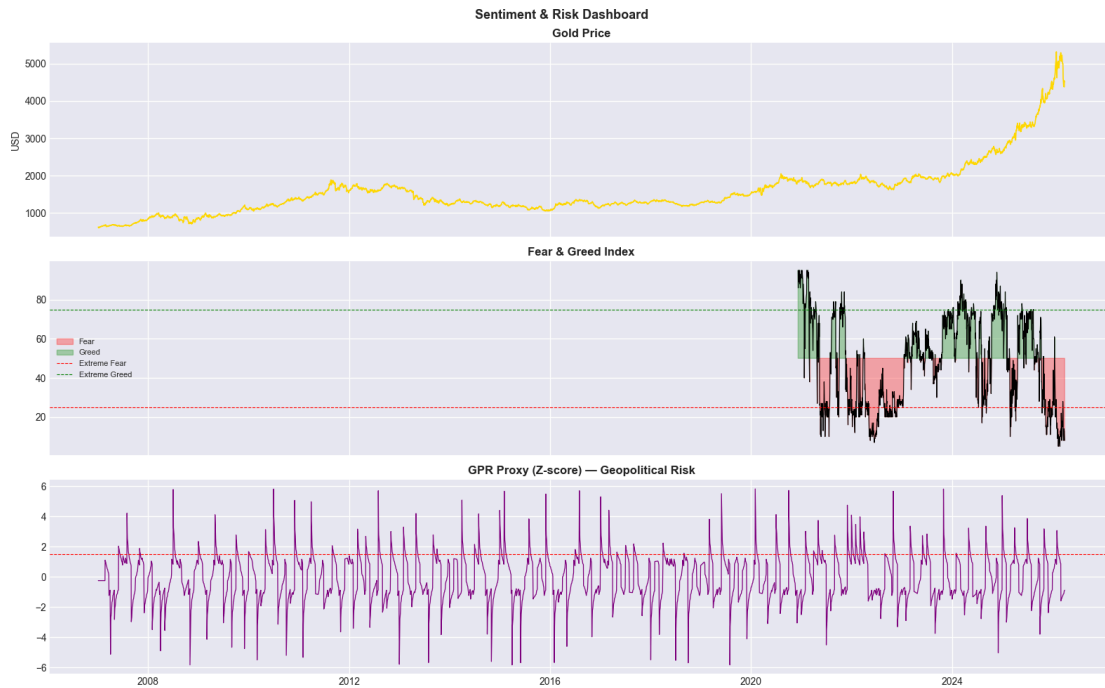


Figure 10. Sentiment and Risk Dashboard Based on Gold Price, Fear-and-Greed Proxy, and GPR Proxy.

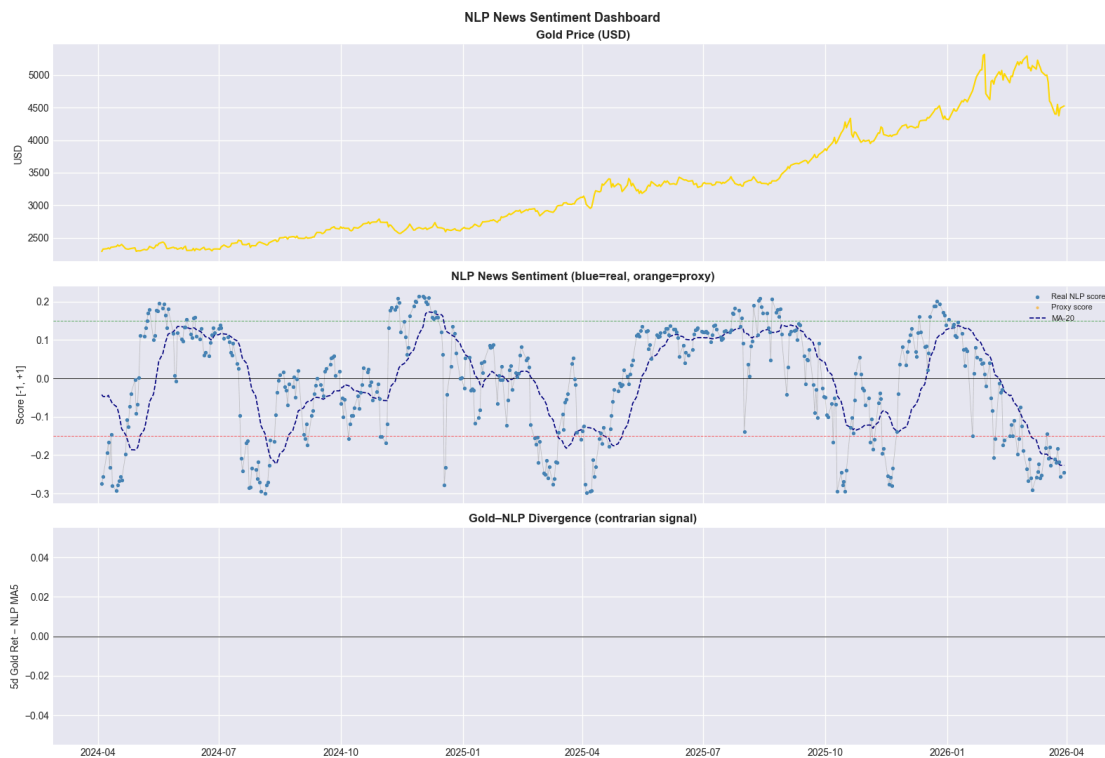


Figure 11. NLP Sentiment Proxy Dashboard and Gold-Risk Divergence Diagnostics.

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GitHub repository link for our research

<https://github.com/emi-lex/wqu-capstone-gold-price-prediction>